

Computer Science & IT

Data Structure & Programming



Array 01

Lecture No. 01



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Topics to be Covered



Topic

1-D array

Topic

2-D array Row major order

Topic

2-D array column major order

Topic

Topic



Topic : Syllabus



Linear Data structure

Array, stack, Queue, **Linked list**

Non Linear Data structure : forest, tree, BST, AVL tree

Hierarchical

Hashing & Graph

Linked List : programming

folder structure

Non Linear





Topic : Syllabus



Data structure is about data organization

& Management

Storage format that enables

efficient Access of data & efficient performance of operation

Time Complexity



Topic : Definition



Array is Collection of similar data type

Stored in Sequential manner.

Size of
data

int : 4B

char : 1B

float : 4B

Base Address : Starting Address of array

Lower Bound : Index of first element

Upper Bound : Index of Last element

C - Language

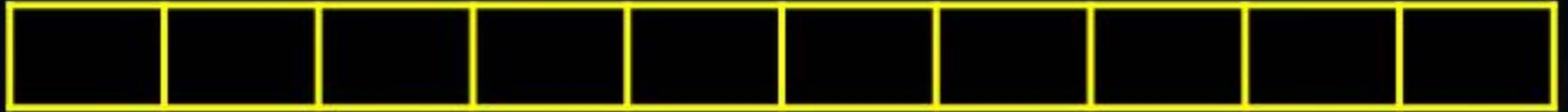
LB is 0



Topic : Question



#Q. Consider the above array $A[1...100]$. Base address of the array is 1000 and each element occupies 2 Bytes of space. What is the address of $A[78]$?



No. of element arranged: $(78 - 1) = 77$

Address: $1000 + 77 \times 2 = \underline{1154}$



Topic : Array



A[1]	A[2]	A[3]	A[4]						
------	------	------	------	--	--	--	--	--	--

1000 1004 1008 1012

Computation Based on No. of element arranged

Before 4, No. of element arranged: $(4 - LB)$

$$\therefore (4 - 1) = 3$$

$$BA + (i - LB) \times \text{Size}$$

$$1000 + (4 - 1) \times 4B = 1000 + 12 = 1012$$



Topic : Question



#Q. Consider the above array $A[0...99]$. Base address of the array is 1000 and each element occupies 2 Bytes of space. What is the address of $A[78]$.

No. of element ~~crossed~~ : $(78 - 0) = 78$

$$BA + (i - LB) \times \text{Size}$$

$$1000 + 78 \times 2 = \underline{1156}$$

No. of element

Question A $\begin{matrix} [-10 \dots 29] \\ \text{LB} \quad \text{UB} \end{matrix} \Rightarrow \text{UB} - \text{LB} + 1 = 29 - (-10) + 1 = 40$

if No. of element in array is x and

No. of element arranged before $A[-1]$ is y

$$\begin{matrix} (i - \text{LB}) \\ = (-1 - (-10)) = 9 \end{matrix}$$

then $x - y$ is $40 - 9 = 31$

$A[LB \dots UB]$

No. of elements is $UB - LB + 1$

$$\begin{array}{c} 1 \quad \text{---} \quad 5 \\ \quad \diagdown \quad \diagup \\ \text{Inclusive} \end{array} = 5 - 1 + 1 = \underline{5}$$

No. of element arranged before $A[i]$

$$(i - LB)$$



Topic : 2 D Array



2- Dimensional Array



Topic : 2 D Array

$A[1..4][1..4]$
↑ Row ↑ column

	lb_2		ub_2	
lb_1	A[1][1]	A[1][2]	A[1][3]	A[1][4]
.....	A[2][1]	A[2][2]	A[2][3]	A[2][4]
	A[3][1]	A[3][2]	A[3][3]	A[3][4]
ub_1	A[4][1]	A[4][2]	A[4][3]	A[4][4]

1. Row major order

2. Column major order

Sequential Arrangement															
Row-1				Row-2				Row-3				Row-4			
A_{11}	A_{12}	A_{13}	A_{14}	A_{21}	A_{22}	A_{23}	A_{24}	A_{31}	A_{32}	A_{33}	A_{34}	A_{41}	A_{42}	A_{43}	A_{44}
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018	1020	1022	1024	1026	1028	1030





Topic : 2 D Array

	lb_2		ub_2	
lb_1	A[1][1] ✓	A[1][2] ✓	A[1][3] ✓	A[1][4] ✓
.....	A[2][1]	A[2][2]	A[2][3]	A[2][4]
	A[3][1]	A[3][2]	A[3][3]	A[3][4]
ub_1	A[4][1]	A[4][2]	<u>A[4][3] ✓</u>	A[4][4] ✓

$A[4][3]$

Before 4th Row, No. of Rows arranged : $(4 - LB) = (4 - 1) = 3$

No. of element in each Row : 4

total : $3 \times 4 = 12$

In 4th Row we are in 3rd column

No. of elements arranged : $(3 - 1) = 2$

total : $12 + 2 = 14$

Address :

$$1000 + 14 \times 2 = 1028$$



Topic : 2 D Array

	lb_2		ub_2	
lb_1	A[1][1]	A[1][2]	A[1][3]	A[1][4]
.....	A[2][1]	A[2][2]	A[2][3]	A[2][4]
	A[3][1]	A[3][2]	A[3][3]	A[3][4]
ub_1	A[4][1]	A[4][2]	A[4][3]	A[4][4]

$A[1..4][1..4]$

of elements in Row
Size of column

of elements in column
Size of Row



Topic : Question

#Q. Consider the following array $A[1..100][1..100]$. Base address is 100 and each element occupies 2-Bytes of space. What is the address of $A[50][49]$ in row major order?

$$\begin{aligned} & 100 + \left[(50-1) \times (100-1+1) + (49-1) \right] \times 2 \\ &= 100 + \left[49 \times 100 + 48 \right] \times 2 \\ &= 100 + 4948 \times 2 \\ &= 100 + 9896 = 9996 \end{aligned}$$



Topic : 2 D Array



$$A[LB_1 \dots UB_1][LB_2 \dots UB_2]$$

Address of $A[i][j]$, BA is Base Address. Size is S

$$BA + \left[\underbrace{(i - LB_1)}_{\substack{\# \text{ of Rows} \\ \text{crossed}}} \times \underbrace{(UB_2 - LB_2 + 1)}_{\substack{\# \text{ of element in} \\ \text{each Row}}} + \underbrace{(j - LB_2)}_{\substack{\# \text{ of element crossed} \\ \text{before } j\text{th column}}} \right] \times S$$

Question: Consider array $A[6..29][-20..-15]$

if No. of elements in Row is $x : (-15 - (-20) + 1) = \underline{6}$

No. of elements in column is $y : (29 - 6 + 1) = 24$

No of element in 2-D array is $z : 24 \times 6 = 144$

then $x + y + z$ is 174
 $6 + 24 + 144 =$

Column Major Order

In Column major order elements are arranged column wise.
Addresses are given below each element.

	lb_2		ub_2	
lb_1	$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$
	$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$
	$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$
ub_1	$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

A_{11}	A_{21}	A_{31}	A_{41}	A_{12}	A_{22}	A_{32}	A_{42}	A_{13}	A_{23}	A_{33}	A_{43}	A_{14}	A_{24}	A_{34}	A_{44}
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018	1020	1022	1024	1026	1028	1030

Column Major Order



$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$
$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$
$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$
$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

$A[2][4]$

No. of columns arranged: $4 - 1 = 3$

No. of element in each column: 4

total: $3 \times 4 = 12$

4th column we are in Second Row

No. of element arranged: $2 - 1 = 1$

total: $12 + 1 = 13$

Address: $1000 + 13 \times 2 = 1026$

Column Major Order



lb_1	$A[1][1]$	$A[1][2]$	$A[1][3]$	$A[1][4]$
i	$A[2][1]$	$A[2][2]$	$A[2][3]$	$A[2][4]$
	$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[3][4]$
ub_1	$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

ub_2

$A[LB_1 \dots UB_1][LB_2 \dots UB_2]$ Column major order

Address of $A[i][j]$. BA - Base Address, S - Size

$$BA + \left[(j - LB_2) \times (UB_1 - LB_1 + 1) + (i - LB_1) \right] \times S$$

of columns
arranged

of elements
in each column

of element
↑
arranged in j th column



Question : 2-D Array, MSQ

$$93 - (-23) + 1 \quad \text{MSQ} \quad 31 - (-7) + 1 = 39$$

Consider the following array $A[-23 \dots 93] [-7 \dots 31]$. Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

- (A) The number of rows is 117 and each row consists of 39 elements in Array A ✓
- (B) The Array $B[0 \dots 116] [2 \dots 40]$ have same number of elements as the array declared in question ✓

(C) Address of $A[0][5]$ is 3538 row major order $\rightarrow 1000 + [(0 - (-23)) \times 39 + (5 - (-7))] \times 2$

(D) Address of $A[0][0]$ is 3538 row major order

Less than 2818

$$\rightarrow 1000 + [23 \times 39 + 12] \times 2$$

$$1000 + [897 + 12] \times 2 \\ = 1000 + 909 \times 2$$

2818



Topic: Practice Problem

1. Consider a 2D array $\text{arr}[-15 \text{ to } 15][-7 \text{ to } 7]$. Find the address of the location $\text{arr}[10][5]$ if the starting address of the array is 500 and size of each element is 4 bytes. (Suppose, the elements are stored in column-major order)

----- $500 + [(5 - (-7)) \times 31 + (10 - (-15))] \times 4$

$$500 + [12 \times 31 + 25] \times 4$$

$$500 + [372 +] \times 4$$

$$= 500 + 397 \times 4$$

$$= 500 + 1588 = 2088$$



3. Consider the natural numbers from are stored in a 2D array `arr[28 to 39][-3 to 10]`. Find the element present at location `arr[35][10]`. (Suppose, the elements are stored in row-major order) _____ 112 ✓

$$BA = 1, S_{ne} = 1$$

$$1 + [7 \times 14 + 13]$$

$$1 + 111 = 112$$

18

112



Topic: Practice Problem



4. Consider the natural numbers from are stored in a 2D array $\text{arr}[\underline{28} \text{ to } \underline{39}][\underline{-3} \text{ to } 10]$. Find the element present at location $\text{arr}[35][10]$. (Suppose, the elements are stored in Column-major order) _____

Natural No begins from 1.

$$1 + \left[(10 - (-3)) \times (39 - 28 + 1) + (35 - 28) \right] \times 1$$

$$1 + [13 \times 12 + 7] \times 1$$

$$1 + 156 + 7 = 164 \checkmark$$

Sparse Matrix

Sparse matrix is a matrix in which Relatively few entries are Non-zero.

1. Lower triangular Matrix ✓ (1996)
2. Upper triangular Matrix
3. Tridiagonal Matrix.

LTM are Square Matrix Size of Row $\equiv$ Size of column

in which elements above principle diagonals are zero

a_{11}	a_{12}	a_{13}	a_{14}
a_{21}	a_{22}	a_{23}	a_{24}
a_{31}	a_{32}	a_{33}	a_{34}
a_{41}	a_{42}	a_{43}	a_{44}

$$A[i][j] = 0 \quad \underline{j > i}$$

1. Row major order

2. Column major order

LTM are Square Matrix Size of Row $\equiv$ Size of column

Nonzero elements are stored one after another

a_{11}	a_{12}	a_{13}	a_{14}
a_{21}	a_{22}	a_{23}	a_{24}
a_{31}	a_{32}	a_{33}	a_{34}
a_{41}	a_{42}	a_{43}	a_{44}

RMO

$\xleftarrow{\text{Row-1}}$	$\xleftarrow{\text{Row-2}}$	$\xleftarrow{\text{Row-3}}$	$\xleftarrow{\text{Row-4}}$						
a_{11}	a_{21}	a_{22}	a_{31}	a_{32}	a_{33}	a_{41}	a_{42}	a_{43}	a_{44}

Formula for RMO:

THANK - YOU